

**Electro Mechanical  
Systems (EE2010)**

Date: February 24<sup>th</sup> 2025

**Course Instructor(s)**

Mr. Muhammad Abdul Majid

Mr. Usama Faiz

**Sessional-I Exam**

**Total Time: 1 Hours**

**Total Marks: 40**

**Total Questions: 02**

**Semester: SP-2025**

**Campus: Lahore**

**Dept: Electrical  
Engineering**

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

**CLO #1:** Investigate a transformer circuit using Faraday's Law and electric circuit theory

**Q1:**

**[marks 20]**

A 60000VA, 4000/230V single phase distribution transformer has series impedance referred to the primary of  $12 + j45 \Omega$ . The components of the excitation branch referred to the secondary side are  $R_c = 50 \Omega$  and  $X_M = j60 \Omega$ . Assume a secondary current of 200A with a lagging PF of 0.7547 is provided to a load at rated secondary voltage.

- Draw circuit model referred to secondary side ✓
- Investigate the circuit to find out Voltage regulation ✓
- Find out applied primary voltage ✓
- Find out real power input to the transformer ✓

**CLO #1:** Investigate a transformer circuit using Faraday's Law and electric circuit theory

**Q2:**

**[marks 5+5+5+5=20]**

Draw B-H curve for DC excitation for a ferromagnetic material, clearly indicate knee point, saturation portion and linear portion

Name two types of losses that occur in a transformer's core. - *Iron, Copper, Hysteresis, eddy.*

Draw phasor diagram of a transformer for capacitive load

Draw the schematic diagram to perform short circuit test on a transformer.

